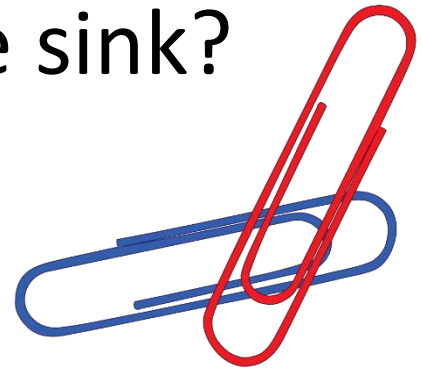
Why Do Some Things Float?

As a class, discuss the following questions:



- Why do some things float and some sink?
- Why can an ocean liner float?
- Why does a paper clip sink?

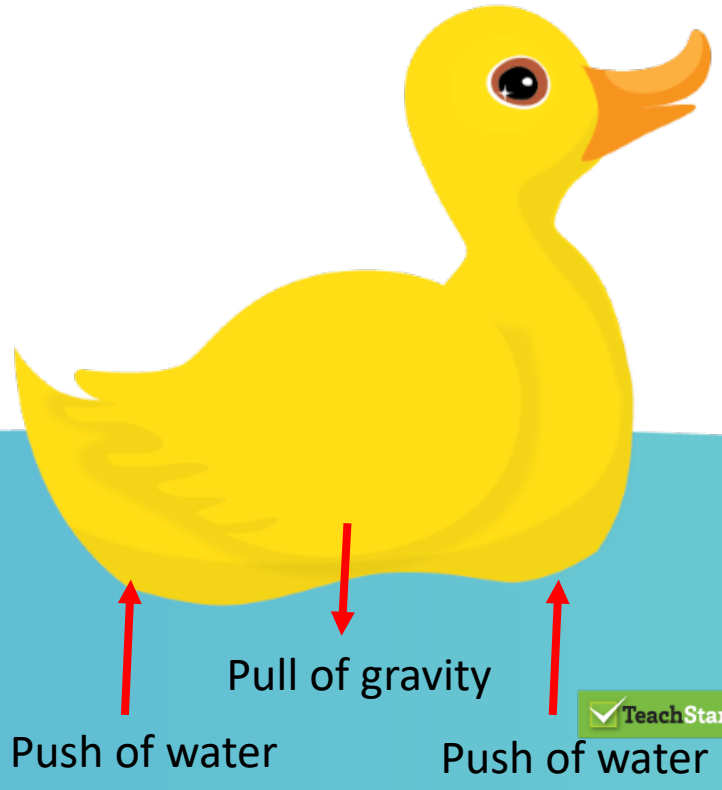


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Buoyancy (Floating)

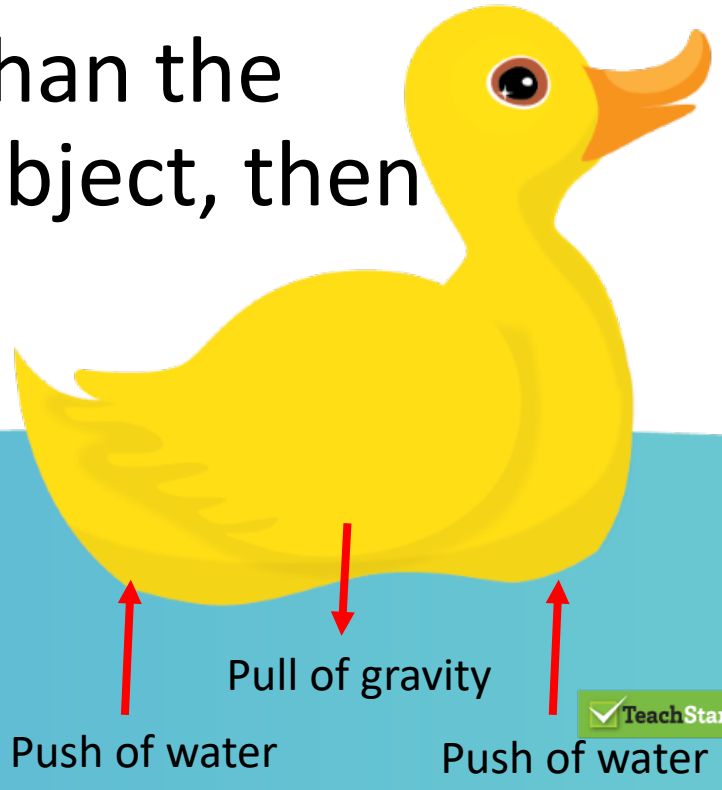
Whether an object floats or sinks is determined by the balance between the downward pull of gravity and the upward push of the water.

This upward pushing force is called buoyancy.



Buoyancy (Floating)

- If the downward pull of gravity and upward push of water are in balance, the object will float.
- If the weight force down is larger than the upward push of the water on the object, then the object will sink.



Investigating Buoyancy

- Put a large ball (a basketball, netball or soccer ball) into a large tub filled with water.
- If you push down on the ball and then let it go, what happens?
- Why do you think this happens?
- Share your ideas.



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Name: _____ Date: _____

Investigating Push and Pull in Water - Observation

1. Do you think the ball is going to float in the tub of water?

   Yes / No

2. When you push on the ball in the tub of water, explain what happens:

   _____

   _____

   _____

3. When you let go of the ball, what happens? Why do you think this happens?

   _____

   _____

   _____

4. Draw a picture of the different forces working on the ball while it is in the tub of water. Use arrows to show the forces.

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